PROJECT AGERE • TIER 8 VERSION 2 (8 GB SIMULATED DEPLOYMENT CAP: 6.40 GB RSS)

Statistical & Mathematical Evaluation of Recent Tier 8 v2 Evidence

Hypothesis Testing, Paired Concordance Analysis, Cluster Bootstrap Intervals & Tradeoff Audits (24–27 October 2026)

COMMIT WINDOW	HEAD COMMIT	PRIMARY PARTITIONS	STATISTICAL STANDARD
24–27 Oct 2026 (184 commits)	2e154cb (main)	Dev24 / Selection52 / Fitting73 / Ext190	Exact McNemar, Bowker, Fisher, Cluster CI

EXECUTIVE STATISTICAL VERDICT

1. Novel Candidate Mechanisms Did Not Achieve Statistically Significant QA Improvement: Across all recent Tier 8 v2 evaluations, none of the newly proposed algorithmic candidates—*crossed-question counterfactual verifier continuation*, *multi-stage iterative program repair*, *source-symbolic quotient repair deduplication*, or *offline table-header/catalog materialization*—achieved a statistically significant QA accuracy improvement at α = 0.05 or passed the predeclared Pareto/no-correct-loss adoption gates.

2. Standard Adaptation & Prompt Engineering Achieved Significant Gains over Unadapted Baselines: Statistically significant improvements occurred exclusively in baseline strengthening: 3B MAS Drafter QLoRA adaptation (0/24 → 11/24, exact McNemar $p = 0.00098$), 3B Binary Critic SFT gate discrimination (12/24 → 23/24 on controlled probe, $p = 0.00098$; 3/24 → 11/24 correct deliveries on natural errors, $p = 0.0078$, though with +4/24 wrong deliveries), and 128-token analysis prompting on 7B SAS (7/24 → 13/24, $p = 0.0313$).

CROSSED VS CONTROL MAS (N=52)

$\Delta_{wrong} = -7.69\%$

Exact $p = 0.2188$ (Non-Sig) • $\Delta_{corr} = 0.0\%$

CROSSED VS ORIGINAL MAS (N=52)

$\Delta_{corr} = -1.92\%$

Loses 1 correct answer • Fails Pareto rule

ITERATIVE REPAIR GAIN (N=52)

$\Delta_{corr} = 0.00\%$

$p = 1.0000$ • SAS adds +2 wrong (+19.1 min)

3B DRAFTER & CRITIC SFT (N=24)

$\Delta_{acc} = +45.83\%$

Exact $p = 0.00098$ (Highly Significant)

1. Mathematical & Statistical Testing Framework

For paired binary outcomes on N matched evaluation cases with 2×2 transition matrix [[a, b], [c, d]], where b denotes discordant gains (0 → 1) and c denotes discordant losses (1 → 0) with $n_d = b + c$, and for 3×3 trinomial delivery states (*Correct*, *Wrong*, *Withheld*) with transition counts n_{ij}:

1. Exact McNemar (Paired Binomial) Test (H_0: $\pi = 0.5$):

$$p_{exact,2} = \min(1,\ 2\ \Sigma_{k=0..\min(b,c)}\ C(n_d,\ k)\ 0.5^{n_d})$$

2. Asymptotic & Continuity-Corrected McNemar ($\chi^2_{df=1}$):

$$\chi^2 = (b - c)^2 / (b + c), \quad \chi^2_{cc} = (|b - c| - 1)^2 / (b + c)$$

3. Paired Risk Difference (Δ) & Wald 95% CI:

$$\Delta = (b - c)/N, \quad SE(\Delta) = N^{-1}\ \sqrt{[\ (b + c) - (b - c)^2/N\ ]}$$

4. Bowker Symmetry & Stuart–Maxwell (3×3 Trinomial):

$$Q_{Bowker} = \Sigma_{i<j}\ (n_{ij} - n_{ji})^2\ /\ (n_{ij} + n_{ji}), \quad Q_{SM} = \mathbf{d}^T\ \mathbf{V}^{-1}\ \mathbf{d}$$

2. Matched Natural MAS Evaluation (N = 52 Cases, G = 15 Reports)

All three MAS critic conditions evaluate the **exact same continued 3B actor** initial drafts, prompts, and plans on 52 report-disjoint selection questions (`evidence/matched-mas-selection-paired-01.json` , `evidence/matched-mas-original-selection-01.json`). Ungated initial candidate accuracy is fixed at **4/52 (7.69%)** reported and **6/52 (11.54%)** raw execution.

Critic Condition (with Identical Common 3B Actor)	Reported Metric (Correct / Wrong / Withheld)	Raw Execution Metric (Correct / Wrong / Withheld)	Precision When Delivered (Reported / Raw)	Prompt / Gen Tokens	Inference Time (s)	Peak Tree RSS (GB)
Matched Control Critic (48 updates, mutation negs)	`3 / 19 / 30`	`5 / 17 / 30`	13.64% / 22.73%	162,851 / 4,662	2,050.7 s	5.060 GB
Crossed-Question Critic (48 updates, crossed negs)	`3 / 15 / 34`	`5 / 13 / 34`	16.67% / 27.78%	170,007 / 4,900	2,136.1 s	5.067 GB
Stronger Original Critic (393 updates, baseline)	`4 / 22 / 26`	`6 / 20 / 26`	15.38% / 23.08%	153,639 / 4,492	1,929.1 s	< 6.400 GB

Paired Comparison & Metric (N = 52, G = 15)	Discordant (b, c)	Paired $\hat{\Delta}$ (SE)	Wald 95% CI	Report-Cluster Bootstrap 95% CI	Exact McNemar $p_{2\text{-sided}}$ ($p_{1\text{-sided}}$)	χ^2 (p) / χ^2_{cc} (p_{cc})	Bowker / SM 3×3 Q (p_{active})	Verdict
Crossed vs Control: Correct Delivered	(0, 0)	0.0000 (0.0000)	[0.0000, 0.0000]	[0.0000, 0.0000]	1.0000 (1.0000)	0.00 (1.00) / 0.00 (1.00)	Q = 2.6667 (df=1, p=0.1025) df=3: p=0.4459	ZERO CHANGE
Crossed vs Control: Wrong Delivered	(1, 5)	−0.0769 (0.0459)	[−0.1668, +0.0130]	[−0.1739, +0.0204]	0.2188 (0.1094)	2.67 (0.102) / 1.50 (0.221)		NOT SIG (P=0.22)
Crossed vs Original (Full): Correct Delivered	(0, 1)	−0.0192 (0.0190)	[−0.0566, +0.0181]	[−0.0588, 0.0000]	1.0000 (0.5000)	1.00 (0.317) / 0.00 (1.00)	Q = 4.7692 (df=2, p=0.0921) df=3: p=0.1895	FAILS NO-LOSS GATE
Crossed vs Original (Full): Wrong Delivered	(3, 10)	−0.1346 (0.0668)	[−0.2655, −0.0037]	[−0.2549, −0.0185]	0.0923 (0.0461)	3.77 (0.052) / 2.77 (0.096)		NOT SIG (P=0.09)
Crossed vs Original (Prefix): Wrong Delivered	(3, 9)	−0.1154 (0.0647)	[−0.2421, +0.0114]	[−0.2281, 0.0000]	0.1460 (0.0730)	3.00 (0.083) / 2.08 (0.149)	Q = 4.0000 (df=2, p=0.1353)	NOT SIG (P=0.15)

Post-Hoc Reference-Coverage Stratification & Interaction Test (`evidence/mas-representation-strata-01.json`)

- **Directly Mappable Cases (n = 23):** Control (`3 / 6 / 14`) and Crossed (`3 / 6 / 14`) have identical marginal counts; 1 wrong delivery was prevented and 1 new wrong delivery was introduced (b = 1, c = 1, $\hat{\Delta}$ = 0.0000, exact McNemar p = 1.0000).
- **Missing Catalog Operand Cases (`unretrieved_operand` , n = 16):** Control (`0 / 6 / 10`) vs. Crossed (`0 / 2 / 14`) accounts for all 4 net fewer wrong deliveries (b = 0, c = 4, exact within-stratum McNemar $p_{2\text{-sided}}$ = 0.1250).
- **Cross-Stratum Enrichment (Fisher's Exact Test):** Among the 19 Control wrong deliveries, 4/6 (66.7%) were withheld by Crossed in `unretrieved_operand` vs. only 1/13 (7.7%) in all other strata combined (Fisher exact $p_{2\text{-sided}}$ = **0.0173**, χ^2 = 7.3634, p = 0.0067; across all 52 cases: 4/16 vs. 1/36, Fisher p = **0.0269**). This proves that the crossed critic's gate shift is post-hoc associated with missing catalog operands rather than improved semantic verification on directly mappable programs.

3. Multi-Stage Iterative Repair & Bounded Local Repair Pool Audits

Across all completed natural selection trajectories (`N = 52`) and development pilots (`N = 24`), we test whether the repair stage (or bounded local repair pool) improves accuracy relative to the no-repair initial prefix:

System / Repair Ablation	N	Repair Attempts / Changed Programs	Discordant Correct (Gains b / Losses c)	Discordant Wrong (Added b / Removed c)	$\hat{\Delta}_{correct}$ / $\hat{\Delta}_{wrong}$	Exact McNemar *p* (Correct / Wrong)	Extra Compute Overhead (Tokens & Wall Time)	Statistical Verdict
Control MAS: Full vs No-Repair Prefix	52	30 / 11 changed	(0, 0)	(0, 0)	0.0000 / 0.0000	1.0000 / 1.0000	+64,025 prompt / +1,555 gen (+786.4 s = 13.1 min)	EXACT ZERO GAIN
Crossed MAS: Full vs No-Repair Prefix	52	34 / 12 changed	(0, 0)	(0, 0)	0.0000 / 0.0000	1.0000 / 1.0000	+71,181 prompt / +1,793 gen (+875.0 s = 14.6 min)	EXACT ZERO GAIN
Original MAS: Full vs No-Repair Prefix	52	27 / —	(0, 0)	(1, 0)	0.0000 / +0.0192	1.0000 / 1.0000	+54,813 prompt / +1,385 gen (+670.1 s = 11.2 min)	+1 WRONG DELIVERY
Control SAS (7B): Full vs No-Repair Prefix	52	23 / —	Deliv: (0, 0) Cand: (0, 2) rep / (0, 3) raw	Deliv: (2, 0) (EW/2004 ×2)	Deliv: 0.0 / +0.0385 Cand: −0.0385 / −0.0577	Deliv Wrong: 0.5000 Cand Raw: 0.2500	+45,486 prompt / +972 gen (+1,144.8 s = 19.1 min)	+2 WRONG / −3 CAND
Proof-Feedback Repair Pilot vs Plain Retry	24	2 eligible / 3 calls	(0, 0)	(0, 0)	0.0000 / 0.0000	1.0000 / 1.0000	+3,582 prompt / +83 gen (+82.5 s)	EXACT ZERO GAIN
Source-Quotient Local Pool (4-check / 8192-tok)	24	370 cand / 4 groups	Oracle bound: (0, 0)	—	0.0000 (all 4 controls)	1.0000	1/24 cases changed top-4 rep	ZERO BUDGET HEADROOM

4. Natural Control SAS (7B) vs. Natural MAS (3B) on Selection52 (N = 52)

Natural Control SAS completed all 52 cases at 5.356 GB peak tree RSS (`evidence/matched-control-sas-natural-complete-01.json`), yielding `8 / 23 / 21` (reported) and `10 / 21 / 21` (raw) after full repair, or `8 / 21 / 23` (reported) and `10 / 19 / 23` (raw) on the no-repair prefix, with **10/52 (19.23%)** reported and **13/52 (25.00%)** raw ungated initial candidate matches. Because Crossed SAS is still running, we evaluate Control SAS against the completed MAS conditions on the same 52 questions (reporting both unpaired two-proportion/Fisher tests and exact paired McNemar bounds across all mathematically possible case-overlap configurations):

Cross-Architecture Comparison on Selection52 (N = 52)	SAS Count (%)	MAS Count (%)	Difference Δ̂ (95% CI)	Two-Prop Z ($p_{2\text{-sided}}$)	Fisher Exact $p_{2\text{-sided}}$	Exact Paired McNemar Bounds [Best-case p, Worst-case p]	Significance (α=0.05)
Initial Ungated Candidate Correct (Reported): SAS vs MAS Actor	10/52 (19.23%)	4/52 (7.69%)	+0.1154 [−0.0139, +0.2447]	Z = +1.7238 (0.0847)	0.1491	b=6,c=0: **0.0313** | b=10,c=4: **0.1796**	MARGINAL / BOUND-DEP.
Initial Ungated Candidate Correct (Raw): SAS vs MAS Actor	13/52 (25.00%)	6/52 (11.54%)	+0.1346 [−0.0116, +0.2809]	Z = +1.7763 (0.0757)	0.1264	b=7,c=0: **0.0156** | b=13,c=6: **0.1671**	MARGINAL / BOUND-DEP.
Full Delivered Correct (Reported): Control SAS vs Control MAS	8/52 (15.38%)	3/52 (5.77%)	+0.0962 [−0.0206, +0.2129]	Z = +1.5942 (0.1109)	0.2008	b=5,c=0: **0.0625** | b=8,c=3: **0.2266**	NOT SIG (P ≥ 0.0625)
Full Delivered Correct (Reported): Control SAS vs Original MAS	8/52 (15.38%)	4/52 (7.69%)	+0.0769 [−0.0450, +0.1988]	Z = +1.2277 (0.2196)	0.3579	b=4,c=0: **0.1250** | b=8,c=4: **0.3877**	NOT SIG (P ≥ 0.1250)
Full Delivered Wrong (Reported): Control SAS vs Crossed MAS	23/52 (44.23%)	15/52 (28.85%)	+0.1538 [−0.0289, +0.3366]	Z = +1.6291 (0.1033)	0.1536	b=8,c=0: **0.0078** | b=23,c=15: **0.2559**	HIGHER WRONG IN SAS

5. Teacher-Forced Continuation Loss Decomposition (SAS 7B & MAS 3B Critic)

Both matched continuations initialized from the exact same full-epoch LoRA checkpoints and trained for identical steps (64 updates / 512 examples for 7B SAS; 48 updates / 384 examples for 3B MAS Critic) with byte-identical validation sets (`evidence/matched-crossed-sas-complete-01.json` , `evidence/matched-crossed-critic-complete-01.json`):

Validation Family & Task	Rows (n)	7B SAS Control Loss	7B SAS Crossed Loss	SAS Relative Δ (Ratio)	3B MAS Control Loss	3B MAS Crossed Loss	MAS Relative Δ (Ratio)
`crossed_validation | critic` (Target family)	192	0.093800	0.030767	**−67.20%** (0.328×)	0.085376	0.038352	**−55.08%** (0.449×)
`mutation_validation | critic` (Single-edit family)	192	0.009672	0.018942	**+95.85%** (1.958×)	0.014434	0.025361	**+75.70%** (1.757×)
`replay_critic | critic` (Old rehearsal pairs)	32	0.004545	0.006232	**+37.12%** (1.371×)	0.022335	0.016940	**−24.15%** (0.758×)
`replay_drafter | drafter` (Fixed actor rehearsal)	32	0.025758	0.022407	**−13.01%** (0.870×)	Common 3B Actor (16 updates, val loss = 0.033928)		
Task-Weighted Aggregate Validation Loss	448 / 416	**0.033459**	**0.017700**	−47.10% (0.529×)	**0.047784**	**0.030709**	−35.73% (0.643×)

Mathematical Tradeoff Analysis: In both SAS and MAS, optimizing on crossed-question negatives reduces teacher-forced loss on the crossed family by 55%–67% (pulling down the weighted aggregate by 36%–47%), but **degrades loss on standard single-edit mutation negatives by +75.7% to +95.8%**. Furthermore, in shared-backbone 7B SAS, changing only critic negative examples altered drafter validation loss (0.025758 → 0.022407) despite identical drafter training examples, confirming cross-task parameter interference.

6. Offline Representation & First-Row Preprocessing Audits (Fitting N = 73)

Diagnostic inspection of natural errors revealed that some FinQA tables place data values in the first row or after colons in headers. Three offline preprocessing experiments were evaluated on the 73 fitting cases (32 reports; 0 neural forwards run yet):

Offline Representation Variant (Fitting N = 73)	Mappable Before → After	Discordant (Recovered b, Lost c)	Paired Δ̂ (SE) [Wald 95% CI]	Exact McNemar $p_{2\text{-sided}}$ ($p_{1\text{-sided}}$)	Asymptotic χ^2 (p) / Edwards χ^2_{cc} (p_{cc})	Catalog Atoms & Extra Prompt Tokens	Statistical Verdict
1. Naive First-Row Top-8 Reranking (`first-row-fitting-test-01.json`)	41/73 → 41/73 (56.16% → 56.16%)	(1, 1)	0.0000 (0.0194) [−0.0380, +0.0380]	1.0000 (0.7500)	0.000 (1.000) / 0.000 (1.000)	8 units fixed (displaced `SNA/2018`)	NULL (P=1.00)
2. Source-Traced Header Materialization (`header-materialization-fitting-01.json`)	41/73 → 45/73 (56.16% → 61.64%)	(4, 0)	+0.0548 (0.0266) [+0.0026, +0.1070]	0.1250 (0.0625)	4.000 (0.0455) / 2.250 (0.1336)	1,922 → 2,105 atoms +5,418 tok (med +72)	EXACT P=0.125 (NON-SIG)
3. Source-Span Catalog-Only Control (`source-span-catalog-fitting-01.json`)	41/73 → 45/73 (56.16% → 61.64%)	(4, 0)	+0.0548 (0.0266) [+0.0026, +0.1070]	0.1250 (0.0625)	4.000 (0.0455) / 2.250 (0.1336)	1,922 → 2,338 atoms +5,597 tok (med +31)	EXACT P=0.125 (NON-SIG)

- **Attribution Proof (`header-visibility-attribution-01.json`):** 183/183 (100.0%) of added quantity occurrences and 240/240 (100.0%) of nonempty first-row cell literals were already present in the original selected source spans. Moreover, Catalog-Only recovers the **exact same 4 cases** (`BLK/2009`, `GRMN/2008`, `IPG/2003`, `VNO/2008`) without modifying source text.
- **Token Cost Sign Test (Catalog-Only vs. Header Materialization):** Catalog-Only uses fewer prompt tokens on 38 cases, equal on 9, and more on 26 (net +179 tokens across 73 cases). Exact two-sided binomial sign test on the 64 non-tied cases: ***p* = 0.1686** (χ^2 = 2.2500, p = 0.1336) — no significant token savings.

7. Baseline Adaptation, Critic Probes & External Evaluations Where Significance WAS Achieved

In contrast to the non-significant novel method candidates above, several foundational baseline and diagnostic evaluations over the 3-day window achieved strong statistical significance:

Experiment & Evidence File	Sample	Comparison & Discordance	Effect Size & 95% CI	Statistical Test & p-Value	Conclusion
MAS 3B Drafter QLoRA Adaptation `finqa-mas-3b-adaptation-dev24-paired-01.json`	N = 24 (Dev24)	Unadapted 0/24 → Adapted 11/24 (b = 11, c = 0)	$\hat{\Delta}$ = +0.4583 Boot CI: [+0.2500, +0.6667]	Exact McNemar ***p* = 0.000977** χ^2 = 11.00 (p = 0.00091)	**SIGNIFICANT (P < 0.001)**
Trained 3B Critic on Controlled Probe `critic3b-controlled-paired-results-01.json`	N = 24 (12 pairs)	Gate: 12/24 → 23/24 (b=11, c=0) Pairs: 0/12 → 11/12 separated	$\hat{\Delta}_{gate}$ = +0.4583 Cluster CI: [+0.3750, +0.5000]	Exact McNemar ***p* = 0.000977** vs Tree (17/24): Fisher p = 0.0479	**SIGNIFICANT (P < 0.001)**
Trained 3B Critic Transfer to Natural24 `critic3b-natural24-delivery-results-01.json`	N = 24 (G = 23)	Correct: 3 → 11 (b=8, c=0) Wrong: 0 → 4 (b=4, c=0)	$\hat{\Delta}_{corr}$ = +0.3333 [+0.167, +0.520] $\hat{\Delta}_{wrong}$ = +0.1667 [+0.040, +0.333]	Correct: Exact ***p* = 0.00781** Wrong: Exact p = 0.1250 Bowker Q = 12.00 (***p* = 0.00248**)	**SIG COVERAGE / TRADEOFF**
SAS 7B Prompt Ablations (Dev24) `finqa-sas-analysis-dev24-paired.json`	N = 24 (Dev24)	128-tok Analysis vs Direct: 7 → 13 (b=6, c=0) Demos vs No-Demos: 1 → 7 (b=7, c=1) 256-tok vs 128-tok: 13 → 11 (b=0, c=2)	128-tok: $\hat{\Delta}$ = +0.250 [+0.083, +0.417] Demos: $\hat{\Delta}$ = +0.250 [+0.042, +0.458] 256-tok: $\hat{\Delta}$ = −0.083 [−0.208, 0.0]	128-tok: Exact ***p* = 0.03125** Demos: Exact p = 0.07031 256-tok: Exact p = 0.5000	**128-TOK SIG (P=0.031)**
External MortarBench & FRAMES Baselines `external-baselines-completed-01.json`	N = 90 (G=9) N = 100	Mortar: MAS 26/90 vs SAS 39/90 (b=7, c=20) FRAMES: MAS 1/100 vs SAS 2/100 (b=1, c=2)	Mortar: $\hat{\Delta}$ = −0.144 [−0.239, −0.045] FRAMES: $\hat{\Delta}$ = −0.010 [−0.050, +0.020]	Mortar: Exact ***p* = 0.01916** FRAMES: Exact p = 1.0000	**SAS BEATS MAS ON MORTAR**
Question-Blind Shortcut Audit (Depth-3 Tree) `critic-question-blind-artifact-audit-01.json`	N = 382 (G = 49)	Orig Val: 275/382 (71.99% vs 50% null) Ext Val: 295/382 (77.23% vs 50% null) Crossed Quartets (N=196): 50.00%	Orig: +21.99% over chance Ext: +27.23% over chance Quartets: 0.00% over chance	Orig: Z = 8.60 (***p* = 8.3×10^{-18}**) Ext: Z = 10.64 (***p* = 1.9×10^{-26}**) Quartets: Z = 0.00 (p = 1.000)	**PROVES SHORTCUT REMOVAL**
Quartet Probe Question-Switch Sensitivity `sas7b-joint-quartet-probe-01.json`	N = 32 (16 pairs)	SAS 7B: 31/32 labels, 15/16 Q-switches Orig MAS 3B: 30/32 labels, 14/16 Q-switches Ext MAS 3B: 27/32 labels (4/8 quartets)	SAS switch: 93.75% (vs 50%) Orig MAS switch: 87.50% Ext vs Orig MAS: −9.38%	SAS switch: Exact ***p* = 0.00052** Orig MAS switch: ***p* = 0.00418** Orig vs Ext Fisher: p = 0.4258	**QUESTION-SENSITIVE (P<0.005)**
Shared-Backbone 3B Role Serving Parity `shared3b-serving-parity-complete-01.json`	2 prompts 4 switches	Full-vocab logits & 32-tok continuations Distinct 4.831 GB → Shared 2.571 GB	max |Δ logit| = 0.00000 Δ RSS = −2.260 GB (−46.78%)	Exact Numerical Identity (13,660+ memory samples)	**EXACT PARITY (−46.8% RAM)**